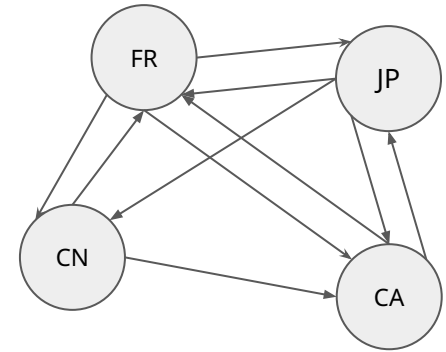


Exploring Structures in IMF Export Network

CSCI 5352
Scott Silverstein

Imf Export Network

Internal Structure



210 Nodes: Countries

25694 Edges: Export Amounts in Millions of Dollars

Dense

- Density: $0.585418 = 25694 / (210^2)$

Not quite Complete

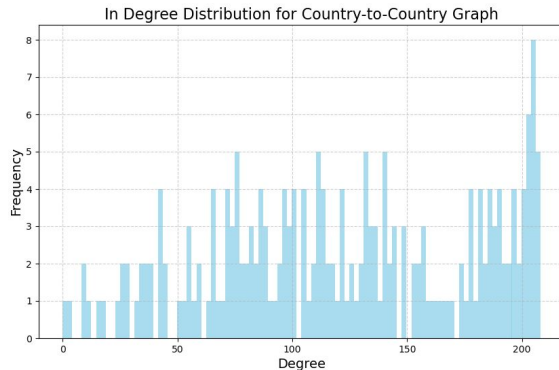
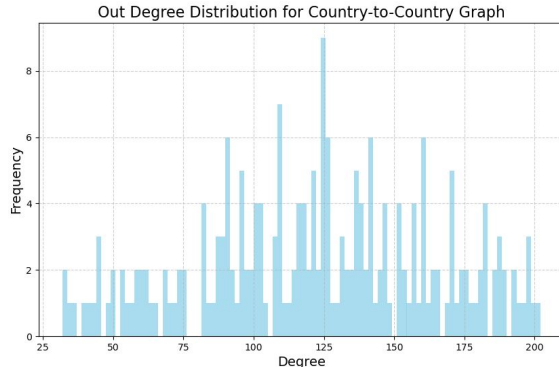
- Each node is connected to many but not all other node.

Directed

- Edges are directed exports from Country A to Country B
- But are directed both ways

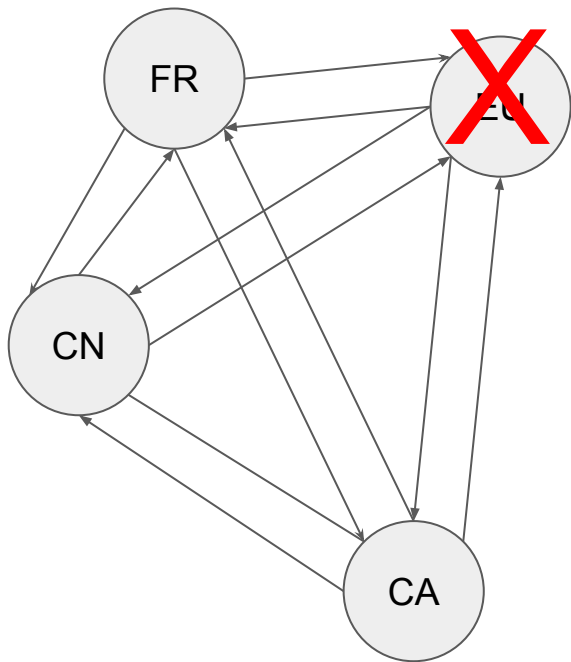
Weighted

- We will look for the structure from this edge data



Data Preparation

Filtering



Nodes

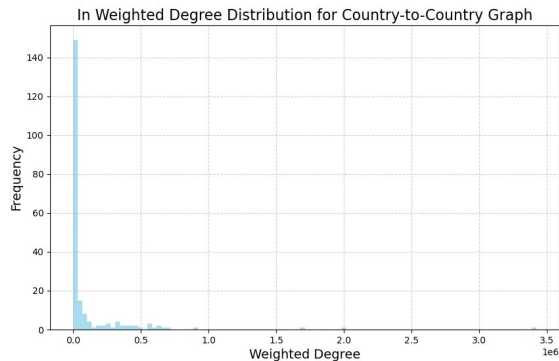
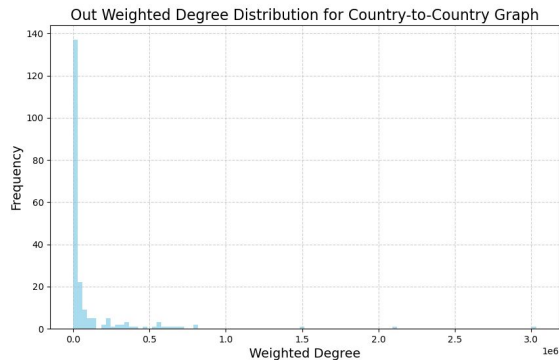
- Filtered out group and continent level data (No Europe group, Asia group, Americas group)
- Country $\leftrightarrow$ country export totals

Year

- Only **2023**

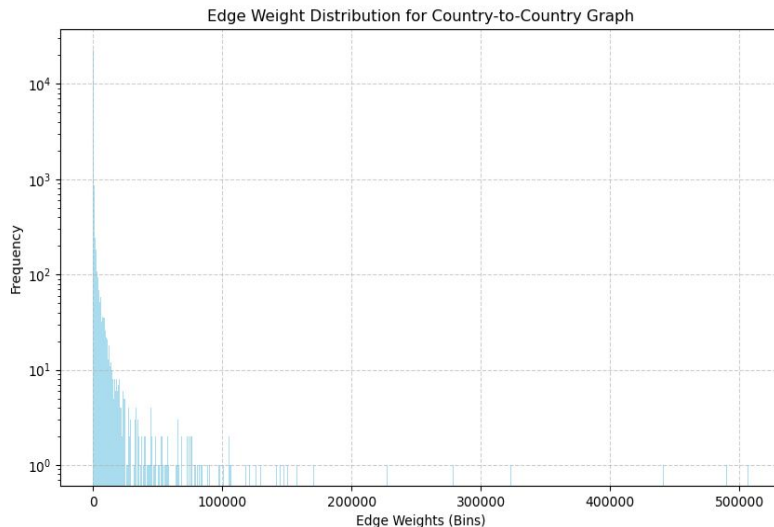
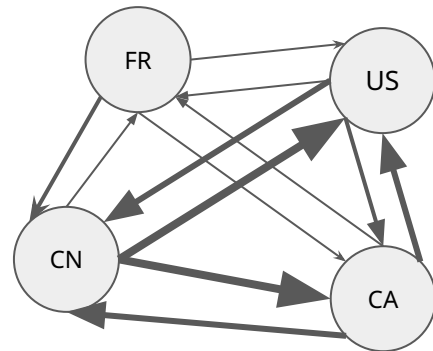
Imf Export Network

Internal Structure



Weighted

- Lots of low weight connections
- Just a few nodes with very high exports



Detecting the structure

SBM vs WSBM

Standard SBM: Models groups based on edge presence/absence.

Requires **thresholding** weighted data

Information Lost

Weighted SBM (WSBM): Models relationships and groups using weights of edges.

- Works on **continuous** data like export amounts.

Goal: Model groups or relationships behind the observed trade volumes.

WSBM Modeling

Choosing a BlockModel

Tool: Graph-tool Python library <https://graph-tool.skewed.de/> by [Tiago P. Peixoto](#)

Models to Compare:

Flat WSBM: Single layer of blocks

Graph-tool functions:

`minimize_blockmodel_dl`,
`BlockState`

Nested WSBM: Hierarchical structure of blocks

Graph-tool functions:

`minimize_nestedblockmodel_dl`,
`NestedBlockState`

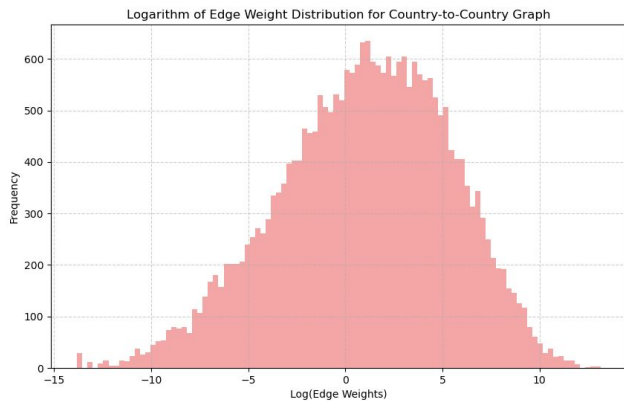
Modeling Parameters:

- `weight-type`: "real-lognormal" | "exponential" Chosen to fit to weight data
- `deg_corr`: Degree Correction - DC-SBM approach, also considers the degree distribution in building the SBM

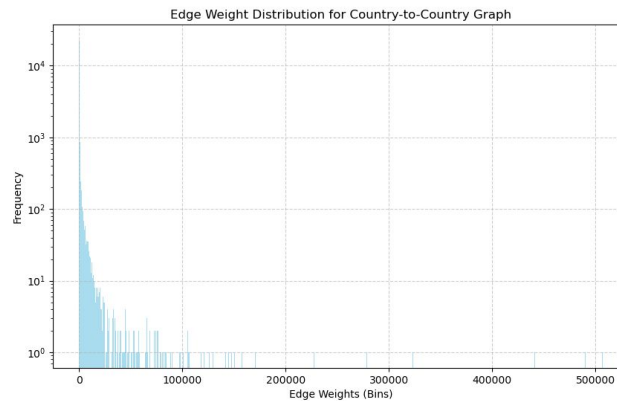
Edge Distribution

Choosing a Covariant Type

What statistical distribution best describes these weights?



Normal Log



Exponential

Fitting the Model

Minimizing Description Length

When calling `minimize_blockmodel_dl` and `minimize_nested_blockmodel_dl`

Process:

- Uses a **Markov Chain Monte Carlo (MCMC)** algorithm that does a guided random walk to converge on the optimal community structure (flat or nested).

The algorithm will **guess** a graph, an error is determined by how well that graph fits and most efficiently explains the hidden structure.

- Minimizes the error **Description Length (MDL)**.
 - For each guess a **MDL** calculated
 - Future guesses will try to reduce this error

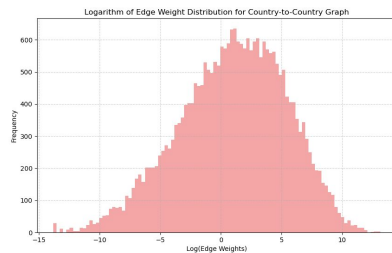
Result: `BlockState` (Flat) or `NestedBlockState` (Nested) object containing the best ranking model

Experiment

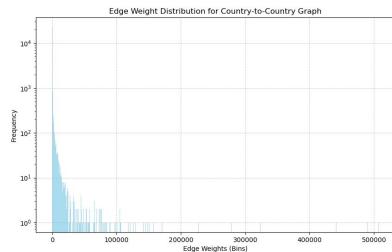
Testing Log Normal vs. Exponential | Nested vs. Flat

Testing: Log Normal vs Exponential

To determine the best fitting weight type
"real-lognormal" and "exponential"



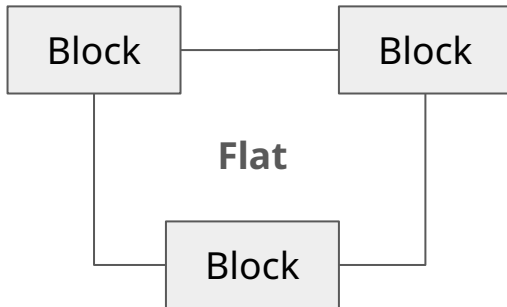
Log Normal



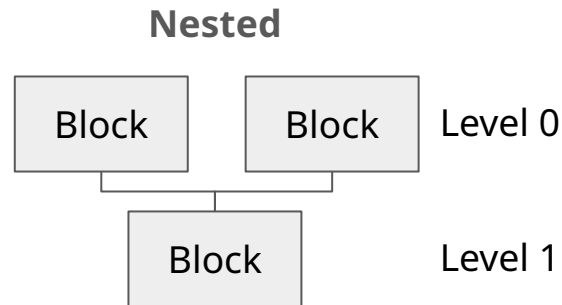
Exponential

Testing: Flat vs Nested

To determine the best fitting structure
minimize_blockmodel_dl and
minimize_nestedblockmodel_dl



Flat



Results

Interpreting Flat Blocks

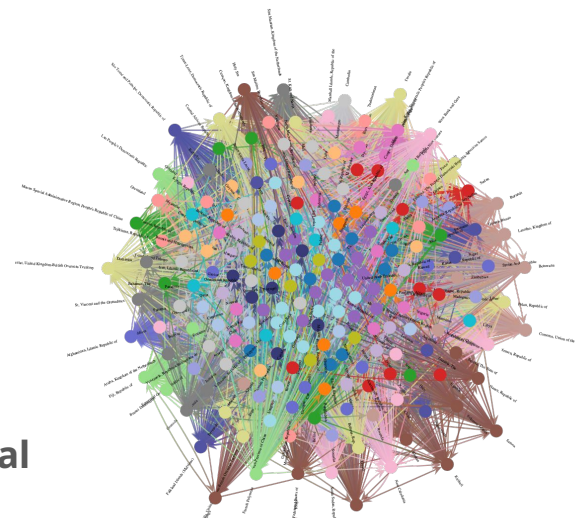
Output: **BlockState** (Single layer of communities).

Flat Exponential DL: 71160.8858

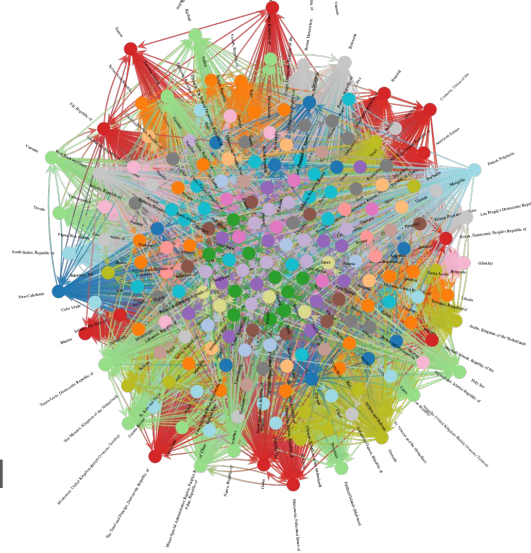
Flat Log-Normal DL: 77913.4790

Exponential was lower

Exponential



Log Normal



Block Memberships (Countries)

Block 12: Croatia, Republic of, Estonia, Republic of, Latvia, Republic of, Lithuania, Republic of, Luxembourg, Romania, Serbia, Republic of, Slovak Republic, Slovenia, Republic of, Ukraine

Block 13: Argentina, Chile, Colombia, Ecuador, Guatemala, New Zealand, Peru, Uruguay

Block 18: Bangladesh, Egypt, Arab Republic of, Israel, Morocco, Pakistan, Saudi Arabia, South Africa

Block 21: Albania, Bosnia and Herzegovina, Iceland, Kosovo, Republic of, Moldova, Republic of, Montenegro, North Macedonia, Republic of

Block 31: Bahamas, The, Belize, Central African Republic, Curaçao, Kingdom of the Netherlands, Haiti, Macao Special Administrative Region, People's Republic of China, Suriname

Block 34: Bolivia, Brunei Darussalam, Fiji, Republic of, French Polynesia, Lao People's Democratic Republic, Papua New Guinea

Block 37: Congo, Democratic Republic of the, Djibouti, Ethiopia, The Federal Democratic Republic of, Kenya, Madagascar, Republic of, Mozambique, Republic of, Namibia, Sudan, Tanzania, United Republic of, Uganda, Zambia

Block 48: Armenia, Republic of, Faroe Islands, Greenland, Kyrgyz Republic, Mongolia, San Marino, Republic of, Tajikistan, Republic of, Turkmenistan

Block 49: Belgium, China, People's Republic of, France, Germany, India, Italy, Netherlands, The, Spain, Switzerland, Türkiye, Republic of, United Arab Emirates, United Kingdom, United States

Block 56: Algeria, Bahrain, Kingdom of, Jordan, Kuwait, Lebanon, Oman, Qatar, Tunisia

Block 58: Bhutan, Eritrea, The State of, Holy See, Kiribati, Micronesia, Federated States of, Nauru, Republic of, Samoa, Solomon Islands, Tonga, Vanuatu

Block 65: Botswana, Burundi, Comoros, Union of the, Eswatini, Kingdom of, Lesotho, Kingdom of, Malawi, Rwanda, Zimbabwe

Block 84: Angola, Burkina Faso, Cameroon, Congo, Republic of, Côte d'Ivoire, Gabon, Ghana, Korea, Democratic People's Republic of, Mauritania, Islamic Republic of, Mauritius, Nigeria, Senegal

Block 86: Chad, Gibraltar, Maldives, Marshall Islands, Republic of the, Nepal, New Caledonia, South Sudan, Republic of, West Bank and Gaza, Yemen, Republic of

Block 113: Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Barbados, Dominica, Grenada, Sint Maarten, Kingdom of the Netherlands, St. Lucia, St. Vincent and the Grenadines

Block 128: Cambodia, Costa Rica, Cuba, Dominican Republic, El Salvador, Guyana, Honduras, Jamaica, Nicaragua, Panama, Trinidad and Tobago, Venezuela, República Bolivariana de

Block 133: Bulgaria, Czech Republic, Finland, Hungary, Ireland, Portugal

Block 142: American Samoa, Anguilla, United Kingdom-British Overseas Territory, Bermuda, Guam, Montserrat, United Kingdom-British Overseas Territory, Palau, Republic of, St. Kitts and Nevis, Timor-Leste, Tuvalu

Block 145: Cyprus, Libya, Malta, Myanmar, Paraguay, Philippines, Sri Lanka

Block 160: Australia, Brazil, Canada, Hong Kong Special Administrative Region, People's Republic of China, Indonesia, Japan, Korea, Republic of, Malaysia, Mexico, Singapore, Taiwan Province of China, Thailand, Vietnam

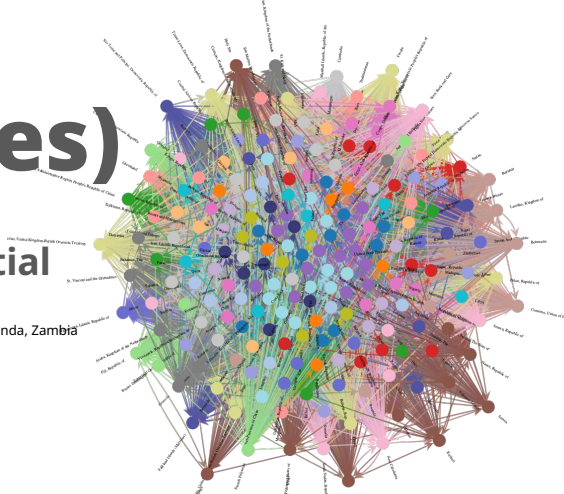
Block 184: Austria, Denmark, Greece, Norway, Poland, Republic of, Russian Federation, Sweden

Block 199: Cabo Verde, Falkland Islands (Malvinas), Gambia, The, Guinea-Bissau, Niger, Sierra Leone, São Tomé and Príncipe, Democratic Republic of

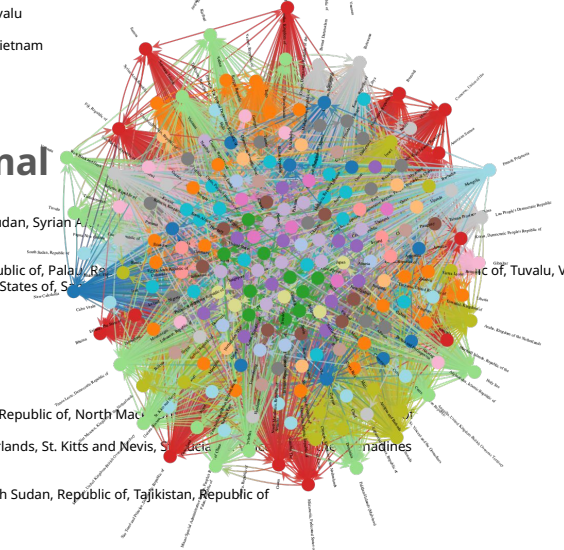
Block 201: Afghanistan, Islamic Republic of, Azerbaijan, Republic of, Belarus, Republic of, Georgia, Iran, Islamic Republic of, Iraq, Kazakhstan, Republic of, Syrian Arab Republic, Uzbekistan, Republic of

Block 205: Benin, Equatorial Guinea, Republic of, Guinea, Liberia, Mali, Seychelles, Somalia, Togo

Exponential



Log Normal



Block 4: Cuba, El Salvador, Guatemala, Guyana, Haiti, Jamaica, New Caledonia, Nicaragua, Panama, Suriname, Venezuela, República Bolivariana de

Block 5: Croatia, Republic of, Estonia, Republic of, Latvia, Republic of, Luxembourg, Serbia, Republic of, Ukraine

Block 6: Angola, Benin, Burkina Faso, Cameroon, Congo, Democratic Republic of the, Congo, Republic of, Djibouti, Gabon, Guinea, Liberia, Mali, Mauritania, Islamic Republic of, Namibia, Rwanda, Sierra Leone, Sudan, Syrian Arab Republic

Block 8: Cambodia, Ethiopia, The Federal Democratic Republic of, Madagascar, Republic of, Mauritius, Mozambique, Republic of, Myanmar, Tanzania, United Republic of, Uganda

Block 22: Belgium, China, People's Republic of, France, Germany, Italy, Netherlands, The, Spain, Türkiye, Republic of, United Kingdom, United States

Block 53: Anguilla, United Kingdom-British Overseas Territory, Falkland Islands (Malvinas), Greenland, Holy See, Kiribati, Lesotho, Kingdom of, Montserrat, United Kingdom-British Overseas Territory, Nauru, Republic of, Palau, Republic of, Samoa, Solomon Islands, Tonga, Vanuatu

Block 64: American Samoa, Bhutan, Burundi, Comoros, Union of the, Eritrea, The State of, Fiji, Republic of, Guam, Korea, Democratic People's Republic of, Marshall Islands, Republic of the, Micronesia, Federated States of, Nauru, Republic of, Palau, Republic of, Samoa, Solomon Islands, Tonga, Vanuatu

Block 70: Argentina, Chile, Colombia, Ecuador, Peru

Block 102: Bulgaria, Czech Republic, Finland, Greece, Hungary, Ireland, Israel, Lithuania, Republic of, Norway, Portugal, Romania, Slovak Republic, Slovenia, Republic of

Block 103: Canada, Hong Kong Special Administrative Region, People's Republic of China, India, Indonesia, Japan, Korea, Republic of, Malaysia, Singapore, Thailand, United Arab Emirates, Vietnam

Block 108: Australia, Brazil, Egypt, Arab Republic of, Mexico, Morocco, Russian Federation, Saudi Arabia, South Africa, Taiwan Province of China

Block 122: Bolivia, Costa Rica, Dominican Republic, Honduras, Paraguay, Trinidad and Tobago

Block 134: Bangladesh, New Zealand, Pakistan, Philippines, Qatar, Sri Lanka

Block 135: Afghanistan, Islamic Republic of, Faroe Islands, Gibraltar, Kyrgyz Republic, Montenegro, San Marino, Republic of, Turkmenistan

Block 142: Albania, Armenia, Republic of, Azerbaijan, Republic of, Belarus, Republic of, Bosnia and Herzegovina, Georgia, Iceland, Iran, Islamic Republic of, Kazakhstan, Republic of, Kosovo, Republic of, Moldova, Republic of, North Macedonia, Republic of

Block 143: Botswana, Brunei Darussalam, Equatorial Guinea, Republic of, Lao People's Democratic Republic, Malawi, Maldives, Mongolia, Nepal, Papua New Guinea, Seychelles, West Bank and Gaza

Block 154: Antigua and Barbuda, Aruba, Kingdom of the Netherlands, Bahamas, The, Barbados, Belize, Bermuda, Curaçao, Kingdom of the Netherlands, Dominica, Grenada, Sint Maarten, Kingdom of the Netherlands, St. Kitts and Nevis, St. Lucia, St. Vincent and the Grenadines

Block 156: Austria, Denmark, Poland, Republic of, Sweden, Switzerland

Block 175: Algeria, Bahrain, Kingdom of, Cyprus, Côte d'Ivoire, Ghana, Iraq, Jordan, Kenya, Kuwait, Lebanon, Libya, Malta, Nigeria, Oman, Senegal, Tunisia, Uruguay

Block 191: Cabo Verde, Central African Republic, Chad, Eswatini, Kingdom of, French Polynesia, Gambia, The, Guinea-Bissau, Macao Special Administrative Region, People's Republic of China, Niger, Somalia, South Sudan, Republic of, Tajikistan, Republic of

Flat Results

Exponential: Winner

Output: BlockState (1 layer of blocks).

Flat Exponential SBM Blocks Found:

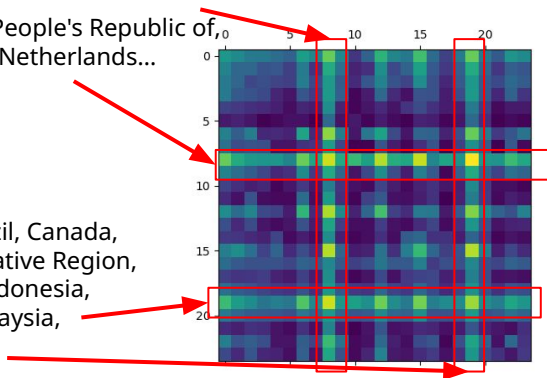
Total: 206

Non-empty: 24

Flat Exponential DL: 71160.8858

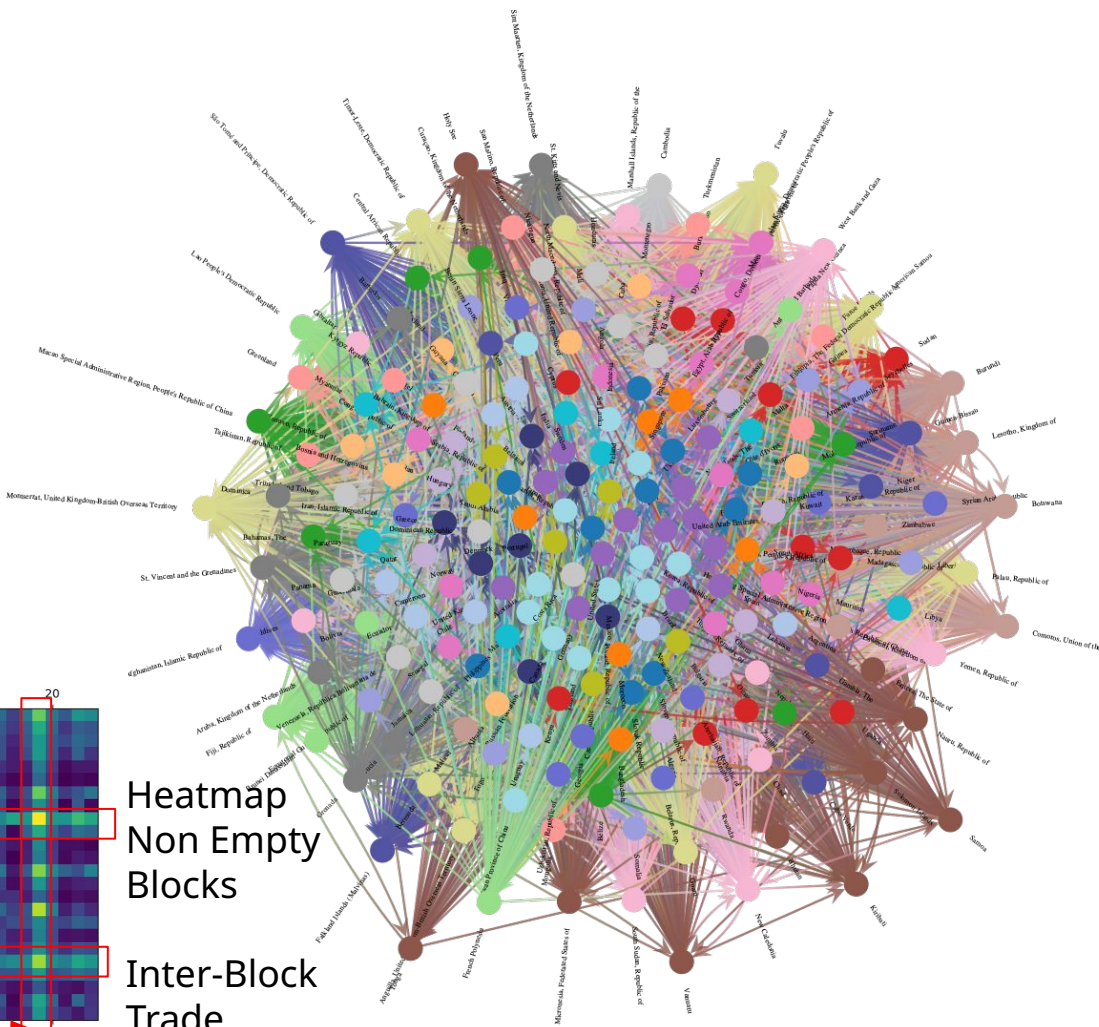
#8 Block 49: Belgium, China, People's Republic of, France, Germany, India, Italy, Netherlands...

#19 Block 160: Australia, Brazil, Canada, Hong Kong Special Administrative Region, People's Republic of China, Indonesia, Japan, Korea, Republic of, Malaysia, Mexico...



Heatmap
Non Empty
Blocks

Inter-Block
Trade



Flat Results

Log Normal

Output: BlockState (1 layer of blocks).

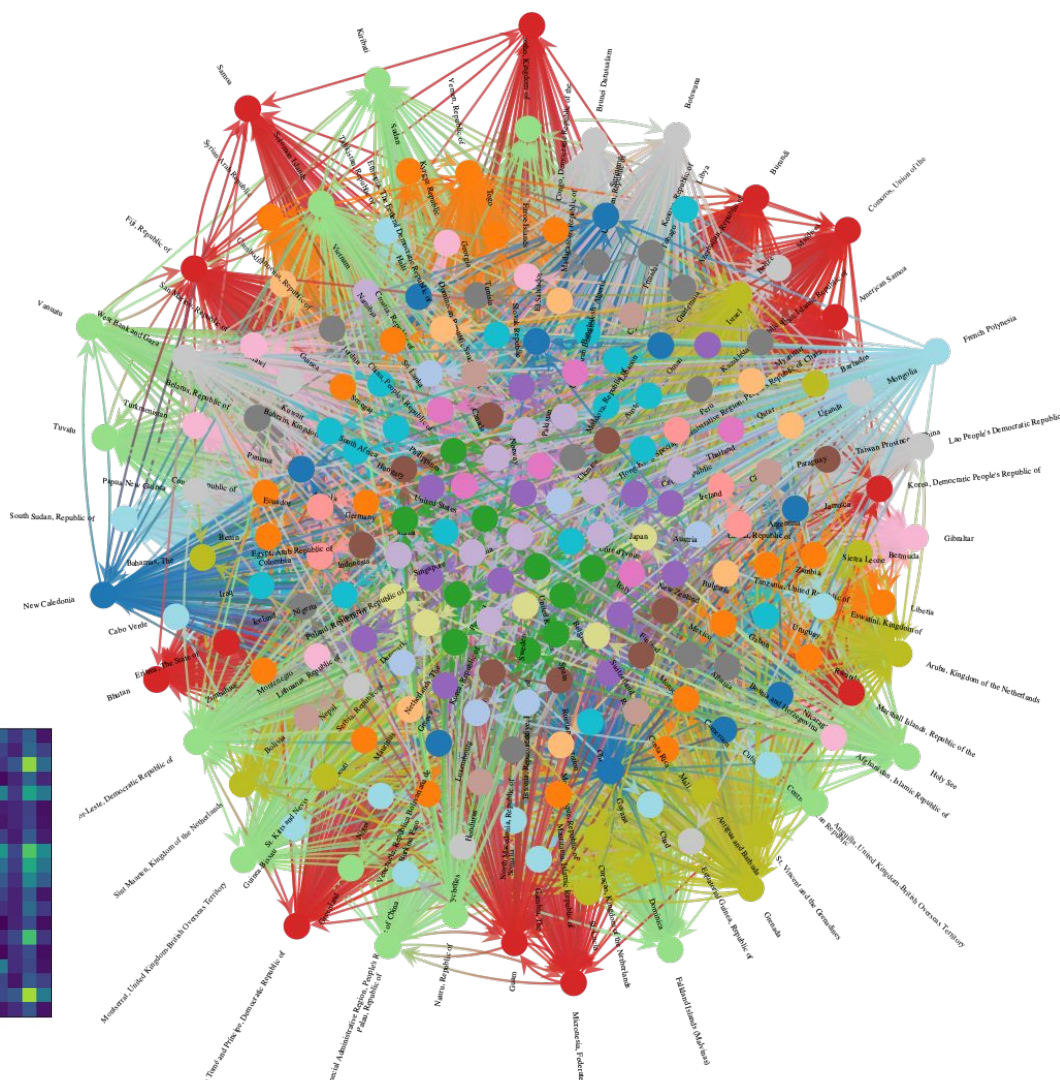
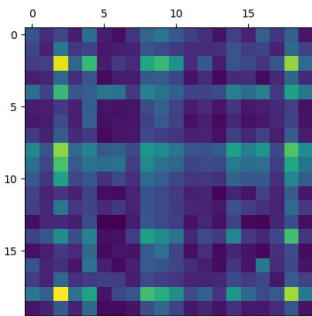
Flat Log-Normal SBM Blocks Found:

Total: 192

Non-empty: 20

Flat Log-Normal DL: 77913.4790

Heatmap
Non Empty Blocks
Inter-Block Trade



Results

Interpreting Nested Blocks

Output: `NestedBlockState` (Levels of Blocks).

Nested Exponential DL: 72408.1542

Nested Log-Normal DL: 79172.5098

Nested Exponential Levels:

Level 0: 205 (10 Non-Empty)

Level 1: 8 (4 Non-Empty)

Level 2: 4 (1 Non-Empty)

Level 3: 1 (1 Non-Empty)

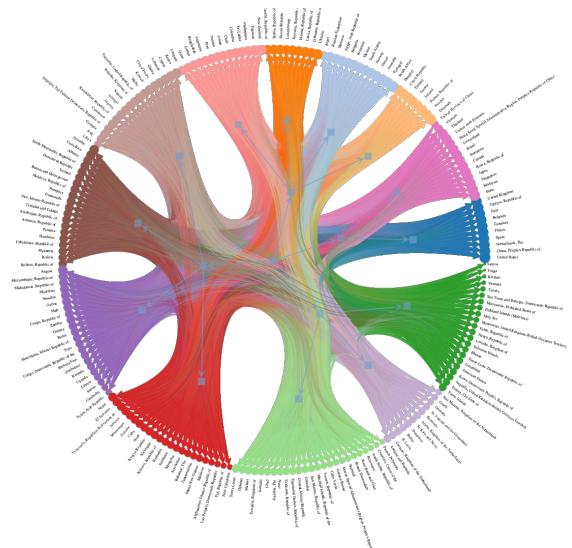
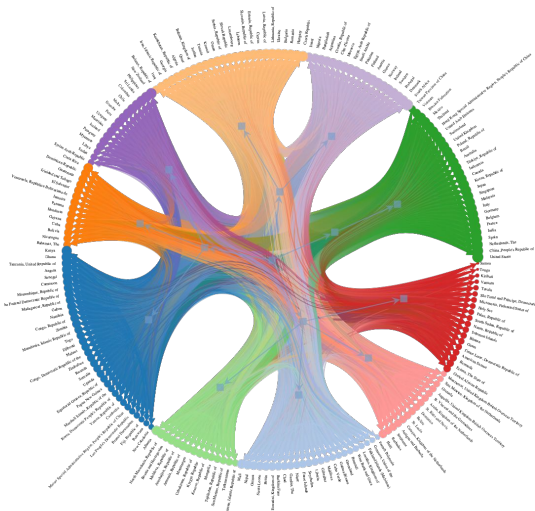
Nested Log Levels:

Level 0: 191 (13 Non-Empty)

Level 1: 11 (4 Non-Empty)

Level 2: 3 (1 Non-Empty)

Level 3: 1 (1 Non-Empty)

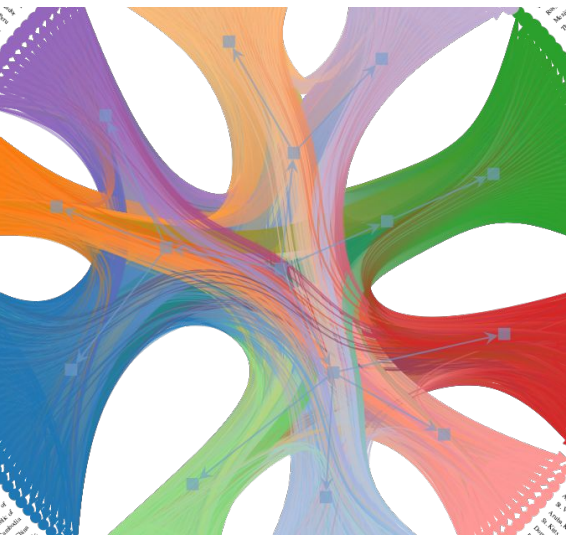


Nested Results

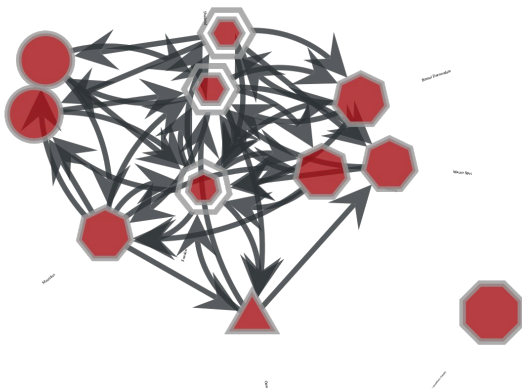
Exponential

Nested Exponential DL: 72408.1542

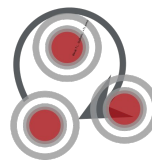
Full



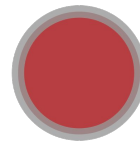
Layer 0: 10 Non Empty
Blocks



Layer 1: 4 Non Empty
Blocks



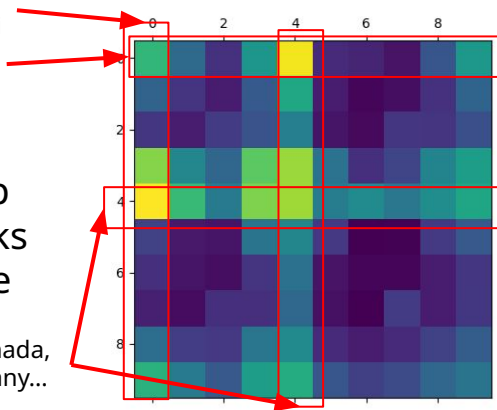
Layer 2: 1
Non Empty Block



0 Block 5: Angola, Botswana, Brunei
Darussalam, Cambodia, Cameroon,
Congo, Democratic Republic of the,
Congo...

Layer 0 Heatmap
Non Empty Blocks
Inter-Block Trade

#4 Block 43: Australia, Belgium, Brazil, Canada,
China, People's Republic of, France, Germany...

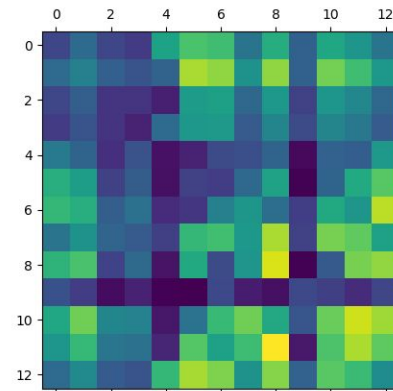


Nested Results

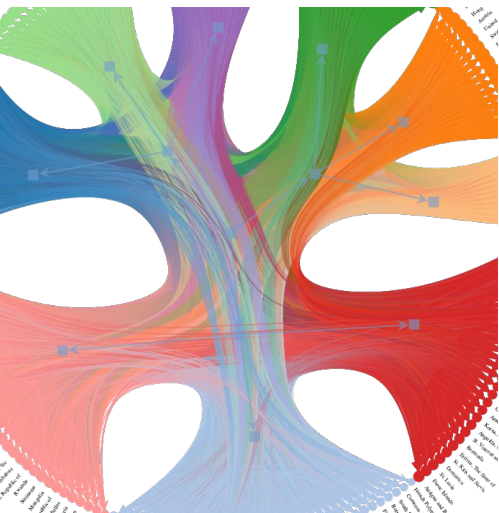
Log Normal

Nested Log-Normal DL: 79172.5098

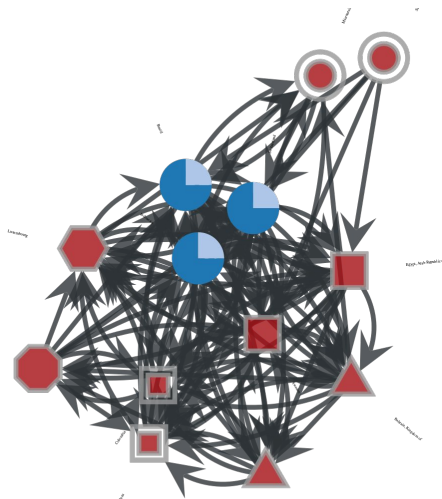
Layer 0 Heatmap
Non Empty Blocks
Inter-Block Trade



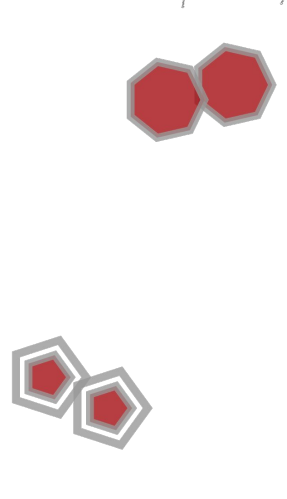
Full



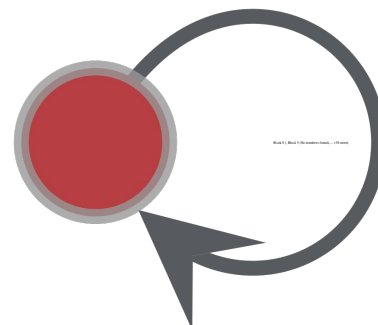
Layer 0: 13 Non Empty
Blocks



Layer 1: 4 Non Empty
Blocks



Layer 2: 1 Block



Conclusions & Summary

WSBM effectively modeled group structure within the network, leveraging trade volumes directly.

Comparing **Flat** vs. **Nested** structures using **Minimum Descriptive Length**, finding the best fit with the least amount of complexity

Flat Exponential DL: 71160.8858

Flat Log-Normal DL: 77913.4790

Nested Exponential DL: 72408.1542

Nested Log-Normal DL: 79172.5098

Flat beat nested in best fit for least amount of complexity

Exponential beat Log-Normal in terms of modeling the distribution of the weights

However, the hierarchical output showed interesting information about potential hierarchical structure within the data. Further study may explore the effects driving this hierarchy.